

For each real-world representation of a context write an inequality that represents the constraint.

1. A child has to be between 30 and 48 inches tall to ride a kiddie ride.

$$30 < x < 48$$

2. A family has a budget of \$400 to spend on groceries each month.

$$0 \leq x \leq 400$$

3. A person has to be at least 16 years old to drive alone in the state of Florida.

$$x \geq 16$$

The function $H(x) = 0.6x + 3.8$ models the height, in inches, of a stack of x cups.

4. Find the value of x for which $H(x) = 10.4$.

$$10.4 = 0.6x + 3.8$$

$$-3.8 \quad -3.8$$

$$\frac{6.6}{0.6} = \frac{0.6x}{0.6}$$

$$x = 11$$

5. In terms of the context, what does the value of x that is the solution to $H(x) = 10.4$ mean?

A stack of 11 cups would yield a height of 10.4 inches.

6. Find the x -intercept.

$$0 = 0.6x + 3.8$$

$$-\frac{3.8}{0.6} = \frac{0.6x}{0.6}$$

$$x = -6.\bar{3}$$

$$(-6.33, 0)$$

7. Determine if the x -intercept is viable. Explain.

The x -intercept is not viable because you can't have a negative number of cups.

8. What key feature is $H(0)$?
The y -intercept
9. The Peace River State Designated Paddle Trail is 67 miles long. Julia is planning on renting a canoe for a canoe trip on the Peace River Paddle Trail. The canoe outfitter charges \$90 to transport the canoe and supplies to the access point. The outfitter charges \$2.50 for each mile of the trail that Julia plans to travel. To plan her budget Julia used the function $C(m) = 2.5m + 90$ where C is the cost in dollars and m is the number of miles she travels.

Select all of the constraints for the context.

- ☒ $m \geq 0$
- ☒ $m \leq 67$
- ☐ $m \leq 90$
- ☐ $C \geq 0$
- ☒ $C \geq 90$
- ☐ $C \leq 257.5$

10. Graph the function in terms of the context.

